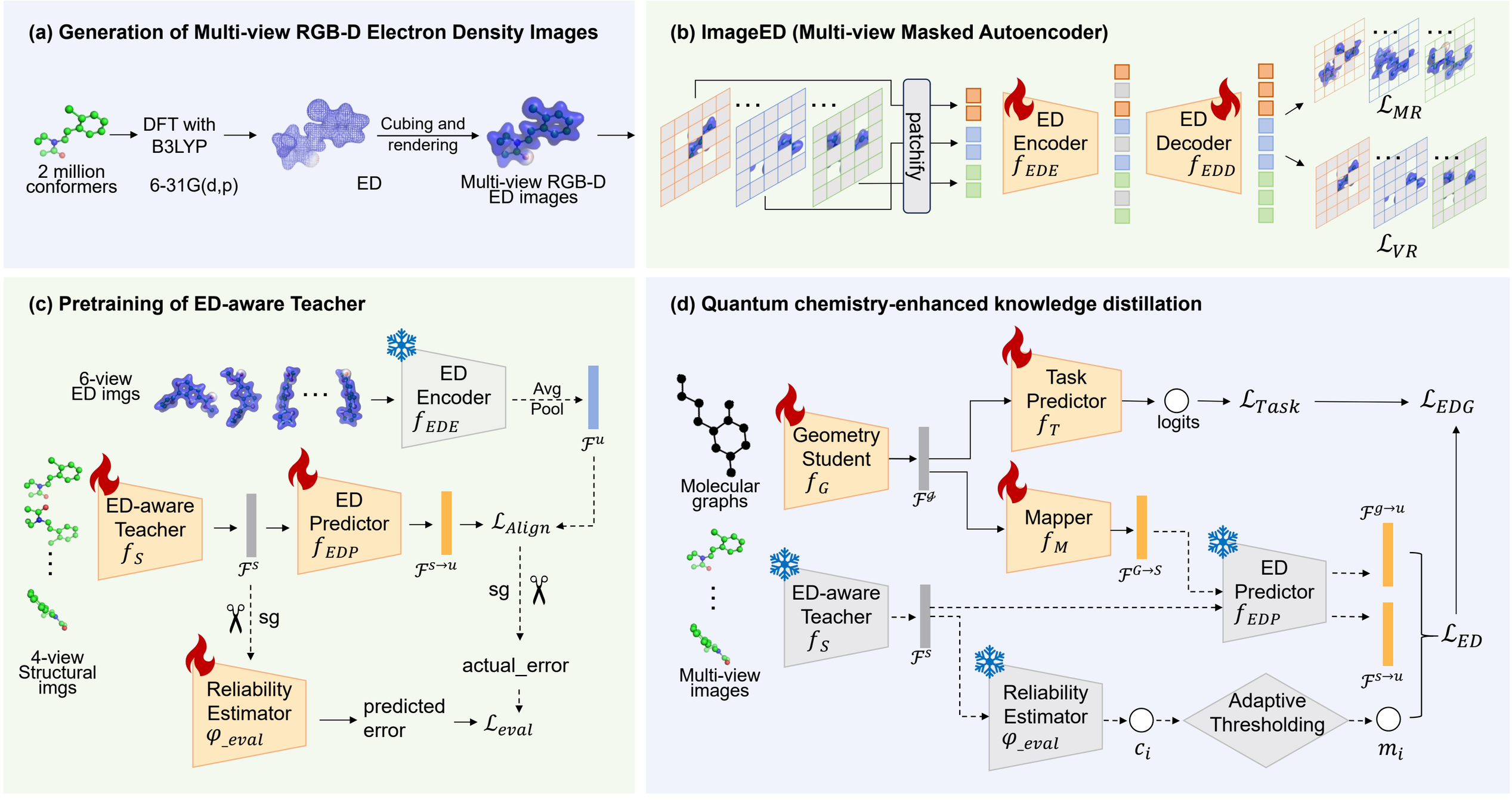
(a) Generation of Multi-view RGB-D Electron Density Images
2 million conformers
DFT with B3LYP
6-31G(d,p)
ED
Cubing and rendering
Multi-view RGB-D ED images
(b) ImageED (Multi-view Masked Autoencoder)
patchify
ED Encoder f_{EDE}
ED Decoder f_{EDD}
$\mathcal{L}_{MR}$
$\mathcal{L}_{VR}$
(c) Pretraining of ED-aware Teacher
6-view ED imgs
ED Encoder f_{EDE}
Avg Pool
$\mathcal{F}^u$
ED-aware Teacher f_S
$\mathcal{F}^s$
ED Predictor f_{EDP}
$\mathcal{F}^{s\to u}$
$\mathcal{L}_{Align}$
sg
4-view Structural imgs
Reliability Estimator φ_eval
predicted error
actual_error
$\mathcal{L}_{eval}$
(d) Quantum chemistry-enhanced knowledge distillation
Molecular graphs
Geometry Student f_G
$\mathcal{F}^g$
Task Predictor f_T
logits
$\mathcal{L}_{Task}$
$\mathcal{L}_{EDG}$
Mapper f_M
$\mathcal{F}^{G\to S}$
$\mathcal{F}^{g\to u}$
ED-aware Teacher f_S
ED Predictor f_{EDP}
$\mathcal{F}^{s\to u}$
$\mathcal{L}_{ED}$
Multi-view images
Reliability Estimator φ_eval
c_i
Adaptive Thresholding
m_i